Diachronic collostructional analysis *

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1 Introduction

This paper discusses a corpus-linguistic approach to meaning change in grammatical constructions – **diachronic collostructional analysis**. Constructions are understood here in the sense of Goldberg (1995, 2006), who defines them as entrenched symbolic units that couple a morphosyntactic form with a meaning. The focus will be on grammatical constructions, as opposed to lexical items, which are also encompassed by Goldberg's definition. Both form and meaning of grammatical constructions are subject to diachronic change. Grammaticalizing forms typically display phonological and morpho-syntactic reduction, and at the same time a semantic shift towards more abstract, grammatical meaning. The present paper focuses on a type of change in grammatical constructions that is less tangible than overt formal change, but that is no less indicative of semantic change. This type of change concerns the interrelations of grammatical constructions and their lexical collocates.

Grammatical constructions are more abstract and schematic than lexical items – not only in meaning, but also in form. While the form of an English noun such as *dog* is phonologically fully specified, the form of a construction like [*BE going to INF*] is not: the construction occurs with different forms of the copula, and it projects a lexical verb in the infinitive, which can be instantiated by just about any verb of the English language. While there is a large amount of variation in the infinitive slot, there are robust tendencies that certain verbs appear more often than others in the construction when their overall text frequency is controlled for. For instance, verbs that are telic and agentive, such as *get* and *put*, occur with *be going to* more often than expected (Hilpert 2008: 112). Assessing the meaning of a construction on the basis of its preferences for certain lexical elements is the main purpose of a collostructional analysis.

If we accept the idea that typical collocates of a grammatical construction reflect its meaning, then it follows that **shifting collocational patterns are indicative of semantic change**. This idea is in fact implicitly held in many grammaticalization studies. For instance, the English auxiliary *can* used to denote a mental ability (Heine 1993: 90), and consequently only took infinitive complements that would express the actions of sentient human beings, as in *teach*, *say*, or *agree*. Over time, as the auxiliary developed semantically, these **selectional restrictions** gradually loosened, and it came to occur with a wider set of collocates. The occurrence of new lexical elements in a construction thus testifies to on-going semantic change. Besides such qualitative changes in the collocate set of a construction, there are also quantitative changes. Certain collocates may become more or less frequent over time. The most frequent collocates of a construction in the 18th century will be different from the most frequent ones in the 19th and 20th century. It was argued in Hilpert (2006, 2008) that a principled, quantitative analysis of shifting collocational preferences could offer some insight into the semantic developments of grammatical constructions.

* This paper draws on materials in Hilpert (2006, 2008) and is intended for teaching purposes. Please refer to my website at <http://omnibus.uni-freiburg.de/~mh608/> or email me at <martin.hilpert@frias.uni-freiburg.de> with any comments or questions.

2 Collostructional Analysis

The basic idea that is presented here is an application of Collostructional Analysis to temporally ordered corpus data. So-called **diachronic corpora** offer comparable sets of texts that represent subsequent periods of time in the development of a language. Amongst many other things, these resources allow us to study the change of collocational patterns over time. For a construction such as *be going to*, we can compare its present-day collocates against its typical collocates in earlier centuries, and this may teach us something about its semantic development. Before we can address the precise methodology of diachronic collostructional analysis, we have to take a closer look at its application to synchronic data.

2.1 Collexeme Analysis

The most basic form of collostructional analysis, **Collexeme Analysis** (Stefanowitsch and Gries 2003), determines which lexical items are most strongly attracted to a given construction and generates a ranked list. This list can be interpreted qualitatively to assess the constructional meaning. A particularly revealing result would be to find that a group of semantically closely related items is attracted to a construction. For instance, a collexeme analysis can show that verbs of transfer (*give*, *send*, *bring*, etc.) are strongly attracted to the English double-object (ditransitive) construction, thus providing support for the idea that this construction carries the meaning of an intended transfer.

To take another example, collexeme analysis can be applied to determine the main verbs that preferably co-occur with a modal auxiliary such as *shall*. The starting point of such an analysis is the **exhaustive extraction** of all examples of *shall* with an infinitive complement from a corpus. From such a collection of examples, all infinitive complements need to be identified and brought into the form of a frequency list. Table 1 shows the twenty most frequent verbs that co-occur with *shall* in the British National Corpus (BNC).

Table 1: Top 20 collocates of *shall*

VERB	TOKENS	VERB	TOKENS
be	2079	consider	225
see	1172	look	225
have	796	tell	213
go	536	get	197
do	466	come	194
say	335	call	172
take	304	put	166
make	251	find	162
give	247	continue	145
return	243	try	142

Actually, many of the verbs that are listed in Table 1 are semantically light and do not tell us very much about the constructional meaning of *shall*. Items such as *be*, *have*, and *do* might further occur frequently with *shall* simply because they are very frequent elements. It is therefore necessary to go beyond raw frequencies. We need to identify those verbs that are not only frequent, but particularly attracted to *shall*. This is done through a comparison of two frequencies. Collexeme analysis determines whether the **observed frequency** of, say *shall*

consider is significantly higher or lower than their **expected frequency**, given how often the elements *shall* and *consider* occur in the corpus. To calculate the expected frequency of the sequence *shall consider*, we need to know how often both individual forms occur by themselves. It is here that reliance on **part of speech tagging** becomes inevitable. The BNC is a tagged corpus, which means that each word form is marked with a category label. This makes it possible to perform an exhaustive search for infinitive verb forms such as *consider*, which look identical to 1st and 2nd person present tense forms. Table 2 presents the twenty most frequent infinitive forms in the BNC.

Table 2: Top 20 infinitives in the BNC

VERB	TOKENS	VERB	TOKENS
be	661,937	think	35,024
have	205,004	give	31,975
do	83,410	find	31,003
get	69,517	come	30,862
see	60,557	want	24,831
go	59,562	like	23,647
make	59,548	help	22,956
know	52,995	look	21,835
take	51,443	tell	21,164
say	41,233	use	20,798

A comparison of Table 1 and Table 2 shows that there is substantial overlap, consisting mainly of highly frequent verbs. Collexeme analysis can filter away this overlap, such that it ranks verbs not by raw frequency, but by their degree of attraction to *shall*. Stefanowitsch and Gries (2003) call the degree of mutual attraction between a construction and a lexical element **Collostructional Strength**. The collostructional strength of *shall* and *consider* is computed through a comparison of the frequencies of these two elements both together and in isolation:

Table 3: Input for a collexeme analysis of *shall* and *consider*

	consider	other verbs	TOTAL
shall	225	15,847	16,072
elsewhere	7,432	3,403,078	3,410,510
TOTAL	7,657	3,418,925	3,426,582

Table 3 should be read as follows: the BNC contains 225 instances of *shall consider*, 16,072 instances of *shall*, and 7,657 instances of *consider*. The corpus as a whole contains some 3.4 million infinitival verb forms. From these pieces of information we can derive the numbers in the outer corners of Table 3. The figures in the remaining five fields are calculated through subtraction. A statistical interpretation of the ratio with which *shall* and *consider* co-occur can tell us whether 225 co-occurrences are unexpectedly many, and if so, how strong the mutual attraction is. To calculate collostructional strength, the four inner fields of Table 3 are submitted to the **Fisher-Yates Exact test**, which indeed finds that 225 instances are disproportionately many and returns a p-value that is very close to 0 (You can run the numbers yourself at <http://www.langsrud.com/fisher.htm >). Put into prose, it is very unlikely that the relatively high frequency of the combination *shall consider* could be due to chance. To obtain a ranked list of attracted verbs, collostructional strength has to be computed for each verb that occurs with *shall* in the BNC. Verbs can then be ordered according to their

p-values, from smallest to largest. Table 4 presents the twenty verbs with the strongest attraction to *shall*, which are referred to as **attracted collexemes**.[1]

Table 4: Top 20 collexemes of *shall*

VERB	COLLSTR	VERB	COLLSTR
see	Infinite	refer	38.45
return	143.46	cease	34.58
consider	103.47	look	28.57
examine	72.47	assume	27.18
call	61.92	endeavor	23.97
discuss	57.94	tell	23.86
apply	52.10	try	23.22
argue	48.08	attempt	22.98
go	47.02	say	21.35
continue	38.52	forget	20.55

In what way can the verbs listed in Table 4 inform a semantic analysis of the English future construction with *shall*? Among the collexemes of *shall*, there is a large set of verbs that have a meta-linguistic, text-structuring function. Explicitly meta-linguistic verbs such as *consider*, *examine*, *discuss*, or *argue* are frequently found with *shall* in statements such as *I shall discuss quantum theory in chapter four*.

Upon inspection of example sentences, it also becomes clear that the two most strongly attracted collexemes *see* and *return* fall into this category. Out of 1,184 examples with *see*, 718 contain the phrase *as we shall see*, often completed by phrases such as *in chapter five*. The verb *return* is used to indicate that a certain topic will be taken up again later. The three collexemes *endeavor*, *try*, and *attempt* are semantically very similar in encoding the intention to carry out an activity. The actual examples reveal that even here a meta-linguistic context is often present, as in *Here we shall try to summarise the main points*, or *I shall attempt to look at these issues in more detail*.

Two separate classes of collexemes encode rules and regulations. The first is instantiated by *call*, and *assume*, which in conjunction with *shall* express the act of a convention, as in *We shall assume that this derivation is a random variable*. Examples with *apply* and *cease* have a similar function, except that they describe a regulation that is made in a contract, such as *Upon the said expiry date our liability shall cease*. Of the 550 examples with *go*, 213 are hortative questions, such as *Shall we go out?* Other examples with *go* encode the intention of an agent, as in *I shall never go to church again*. The collexeme *say* exhibits the same affinity towards interpersonal meanings, as in suggestions such as *Shall we say twelve thirty?* or exclamations such as *What shall I say?*

By uncovering such preferences, collexeme analysis provides a perspective on the individual semantic characteristics of a given construction. Importantly, these tendencies would

[1] The collexemes are listed along with a numerical value of their collostructional strength (COLLSTR).The CollStr value represents the p-value computed by the Fisher-Yates Exact test in an inverse logarithmic function. It is represented in this way for greater ease of recognition, since the p-values tend to become very small for the most attracted elements. In Table 4, a larger value represents a stronger attraction. All of the reported values are statistically significant, as a value of 1.3 or larger represents a p-value of <.05. The value 'Infinite' in the table denotes maximum association strength.

potentially go unnoticed on an approach that considered only raw frequencies. As the English construction with *shall* is very productive and co-occurs with a wide range of complementing main verbs, a collostructional approach is needed to identify verbs with a particular affinity to the construction.

2.2 Distinctive collexeme analysis and its diachronic application

The diachronic application of collostructional analysis is based on a slightly different analytical procedure than the one outlined above, namely **distinctive collexeme analysis** (Gries and Stefanowitsch 2004). This method follows the same logic, but was developed to contrast two or more constructions in their respective collocational preferences. Before turning to its diachronic application, we briefly consider how this method works on the basis of synchronic data.

Distinctive collexeme analysis lends itself in particular to the study of semantically related constructions, as for instance English *will* and *be going to*. Both constructions are used to refer to future events, but besides several other differences, the constructions show distinct co-occurrence patterns with infinitival collocates. Importantly, this difference does not immediately fall out of directly observable raw frequencies. Table 5 lists the ten most frequent verbs in both of these constructions, again based on data from the BNC. As can be seen, there is broad collocational overlap.

Table 5: Top 10 verbs with *will* and *going to* in the BNC

WILL		GOING TO	
VERB	TOKENS	VERB	TOKENS
be	41,947	be	4,756
have	5,906	do	1,907
take	4,150	get	1,403
make	3,182	have	983
do	3,039	take	647
go	2,821	say	643
come	2,732	make	631
give	2,543	go	616
continue	2,477	happen	552
find	2,465	tell	434

In both constructions, the most frequent elements are general verbs that are either semantically light, such as *do*, or polysemous, such as *go*. Distinctive collexeme analysis determines whether there are asymmetries in the relative frequencies of the co-occurring lexical verbs. This is done to highlight those elements that are maximally distinctive for each respective construction. The method identifies all verbs that occur significantly more often with one construction than with the other, and ranks these according to the degree to which they are **distinctive**. Table 6 illustrates the calculation with the example of *say*. The string *will say* actually occurs more frequently than the corresponding string *going to say*, but of course we have to take into account that the construction with *will* is much more frequent than the construction with *be going to*. If we submit the four inner fields of the table to a statistical test, we learn that *say* occurs significantly more often with *be going to* than with *will*.

Table 6: Input for a distinctive collexeme analysis of *say* in *will* and *going to*

	say	other verbs	TOTALS
will	813	185,734	186,547
going to	643	26,294	26,937
TOTALS	1,456	212,028	213,484

The calculation in Table 6 is done for all verbs that are encountered with the respective constructions. The overall result of a distinctive collexeme analysis is a pair of lists presenting the most typical collocates of each construction in ranked order. Table 7 shows the ten most **distinctive collexemes** for both *will* and *be going to* based on data from the BNC.

Table 7: Top 10 distinctive collexemes of *will* and *going to* in the BNC

WILL	COLLSTR	BE GOING TO	COLLSTR
continue	83.57	do	Inf
be	74.17	get	Inf
provide	61.39	say	195.36
include	56.35	happen	135.34
remain	44.76	ask	87.20
receive	42.50	die	78.72
become	41.15	put	74.96
depend	39.41	tell	58.85
enable	37.72	marry	53.99
require	36.58	let	42.95

The two lists in Table 7 can be used to assess the semantic differences between *will* and *going to*. In a comparison of *will* and *going to* based on the ICE-GB, Gries and Stefanowitsch (2004: 114) show that among the distinctive collexemes of *will*, many are non-agentive or low in transitivity. Their results are replicated here with data from the BNC, as Table 7 lists distinctive collexemes of *will* such as *continue*, *include*, *remain*, and *depend*. Conversely, *be going to* has distinctive complements that are agentive and high in transitivity, such as *do*, *say*, *put*, or *marry*. By bringing these elements into focus, distinctive collexeme analysis **accentuates semantic differences** between potentially fairly similar constructions.

Distinctive collexeme analysis can not only contrast two constructions, but it can also be used to compare three or more constructions. The comparison of more than two alternatives lies at the heart of the **diachronic application** of distinctive collexeme analysis. On the basis of diachronic corpora, the method can determine what types of co-occurring elements were typical of a given construction at different historical stages. Unlike the synchronic variant, the diachronic application does not contrast multiple constructions, but studies **a single construction across multiple periods of time**.

For a diachronic application we can stay with the example of English *shall*. First, we search the Penn-Helsinki Parsed Corpus of Early Modern English (PPCEME) to retrieve all instances of *shall* with its orthographical variants and inflected alternatives. The corpus is divided into three periods of time, such that three concordances are obtained. From each collection of examples, the infinitive complements are brought into the form of a frequency list. Table 8 shows for each period the ten most frequent verbs that co-occur with *shall*.

Table 8: Top 10 verbs with *shall* over three periods of time

1500-1570		1570-1640		1640-1710	
Verb	Tokens	Verb	Tokens	Verb	Tokens
be	349	be	372	be	234
have	136	have	151	have	79
see	62	find	80	find	34
come	55	see	66	make	26
find	50	do	65	see	23
make	44	come	62	do	23
do	39	take	52	give	22
take	36	make	45	think	21
go	35	hear	43	forfeit	20
know	31	know	38	take	19

As a tool to investigate the history of *shall*, Table 8 is only of limited use. Many of the most frequent verbs are semantically light, and there does not appear to be much change. This motivates an analysis in terms of elements that are distinctive, not merely frequent.

The **diachronic distinctive collexemes** of *shall* can be determined through a comparison of **raw frequencies** of the type seen in Table 8 with the **overall frequencies** of the compared items. Much as in the synchronic application of distinctive collexeme analysis, the collostructional method abstracts away from items that are common in each period and highlights those that are significantly more frequent than expected, given the totaled frequency of each item across the three periods. Items will be judged as distinctive if they occur frequently in one period but are relatively sparse in the two others. In this way, differences between the three periods are accentuated and the semantic developments become more prominent. Table 9 illustrates the input that is needed to calculate the status of the verb *say* over the three periods. For each period, the table lists the frequency of *say* as compared to all other verbs that occur as complements of *shall*. For example, in the earliest period there are 28 instances of *say* out of a total of 2,138 examples. By comparing the relative frequencies of *shall say* over time, it can be determined accurately whether and how strongly *say* was attracted to *shall* in each period.

Table 9: Input for a diachronic distinctive collexeme analysis of *shall say*

	shall say	*shall* + other verbs	total
1500-1570	28	2,110	2,138
1570-1640	18	2,335	2,353
1640-1710	8	1,169	1,177
total	54	5,614	5,668

The distribution that is represented by the six inner fields of Table 9 is again interpreted with a statistical test in order to see whether it deviates significantly from a chance result. In distinctive collexeme analyses with more than two sets to compare, the applied statistic is not the Fisher-Yates Exact test, but a **Multinomial Test** that allows for more than four values to be compared simultaneously (Gries and Stefanowitsch 2004). The calculation illustrated in Table 9 has to be made for every infinitive complement type that is encountered with *shall* in each of the three periods. Such an analysis yields ranked lists of the most distinctive items for

each period. Table 10 shows the fifteen most distinctive collexemes for each of the three periods ranked by collostructional strength, along with their actual token figures.

Table 10: Top 15 distinctive collexemes of *shall* over three periods of time

1500-1570			1570-1640			1640-1710		
Verb	N	CollStr	Verb	N	CollStr	Verb	N	CollStr
fortune	9	3.80	incur	13	3.38	add	14	5.52
wage	8	3.38	assemble	7	2.68	direct	8	5.46
divide	8	2.60	feed	7	2.68	discover	5	3.41
appear	24	2.07	hear	43	2.56	examine	7	3.39
perceive	11	1.92	offend	12	2.17	stay	6	2.81
understand	26	1.81	fall	14	2.04	refer	4	2.73
beg	7	1.75	contain	7	1.97	stand	11	2.65
require	9	1.66	enjoy	5	1.92	endeavor	7	2.60
say	28	1.61	imagine	5	1.92	extend	15	2.58
attain	5	1.50	do	65	1.79	be	234	2.50
sow	5	1.50	want	15	1.74	seem	11	2.46
show	18	1.43	keep	23	1.70	begin	6	2.42
abide	6	1.43	observe	10	1.64	lead	5	2.25
walk	6	1.43	redeem	4	1.53	export	4	2.11
behold	3	1.27	thirst	4	1.53	administer	4	2.11

The calculations promote the ranking of verbs that are maximally unevenly distributed. As seen above in Table 9, *say* occurs 28 times in the first period, 18 times in the second, and only 8 times in the last. Hence it is judged as distinctive for the first period. The most distinctive verb of the first period, *fortune*, occurs nine times in the first period, but not at all elsewhere. It appears as the most distinctive element of the first period. The most distinctive verbs are therefore those that occur with some frequency, but are **idiosyncratic for just one period**. Changes in the most attracted verbs can be interpreted as a symptom of ongoing semantic change, and the observed tendencies can be compared against **pre-existing hypotheses** about the development of the construction.

A tendency that can be observed in Table 10 is the decline of verbs that encode perception and cognition. The first period lists *perceive*, *understand*, *show*, and *behold*, the second period lists *hear*, *imagine*, and *observe*, and the third lists only *examine*. Conversely, there is an increase of stative verbs over the three periods. The first period lists none, the second period has *contain*, *keep*, and *thirst*, and the third period lists *stay*, *stand*, *extend*, *be*, and *seem*.

While the method of diachronic distinctive collexeme analysis opens up a new perspective on historical corpus data, its **limitations** have to be kept in mind. For instance, the method does not take into account the overall corpus frequencies of the lexical elements that occur with the construction that is investigated. This is an idealization, since most words clearly do not have the same chance of occurrence in different historical periods. Some words become more frequent over time, others cease to be used. Stefanowitsch (2006) further cautions that a claim to **psychological reality** cannot be maintained under a historical application of collostructional analysis. Quite undeniably, the proposed mode of analysis models periods of language change that by far exceed the lifetime of individual speakers. To illustrate the point that Stefanowitsch makes, consider a speaker of English living in the year 1500. Our speaker has no way of anticipating the verbs that will be used frequently with *shall* two-hundred years

later. Consequently, he does not know which of the verbs frequently used with *shall* are distinctive of his own time. Only in hindsight can the distinctive verbs be determined. It would therefore be misguided to interpret the lists in Table 10 as mental representations of the collocational profile of *shall* in the respective period.

3 Collostructional analyses of Dutch *gaan* and English *be going to*

With the general operating principles of collostructional analysis in place, we can now turn to the diachronic development of two Germanic future constructions that derive from verbs of movement. The Dutch lexical verb *gaan* 'go' has grammaticalized into an auxiliary that takes bare infinitive complements and that is used to convey future time reference. In English, the progressive form of the verb *go* combines with the copula *be* and a *to*-infinitive to form the construction *be going to*. Through diachronic distinctive collexeme analyses, we can investigate the collocational preferences of these constructions at different historical stages in order to arrive at an understanding of how their present-day grammatical meaning developed out of the lexical meaning of movement. Both Dutch *gaan* and English *be going to* have been studied by themselves and in comparison to other future constructions in the two respective languages. The following paragraphs survey a few studies that make predictions that are testable through a diachronic distinctive collexeme analysis.

Ten Cate (1991) compares *gaan* against an alternative future construction with *zullen*, finding that the two are dissimilar in several respects. As illustrated by (1a), the construction with *gaan* cannot co-occur freely with stative verbs such as *hebben* 'have' or *zijn* 'be', which unrestrictedly combine with *zullen*. Example (1b) shows that *gaan* is preferred over *zullen* if a future event results from a present cause such as the intention of an agent. This is consonant with the fact that *gaan* can occur as the complement of the modal verb *willen* 'want', which *zullen* cannot, as shown in (1c). Finally, (1d) shows that *gaan* does not convey the epistemic modal interpretations of *zullen*. Both sentences are grammatical, but the variant with *gaan* can express only future time reference. Ten Cate concludes that *gaan* encodes inchoativity and reference to the proximate future, while *zullen* is used contrastively as a marker of more distant future events. His analysis predicts a bias of *gaan* towards dynamic and especially inchoative verbs.

(1) a. Marie [zal / *gaat] [geluk hebben / gelukkig zijn]. (a-d: Ten Cate 1991: 28)
*Mary shall / *goes luck have happy be*
'Mary will be lucky / happy.'

b. Jan en Marie [gaan / *zullen] morgen trouwen.
*Jan and Mary go / *shall tomorrow marry*
'Jan and Mary will get married tomorrow.'

c. Paul wil Duits [gaan / *zullen] leren.
Paul wants German go / shall learn
'Paul wants to learn German.'

d. Paul [gaat / zal] in de badkuip zitten.
Paul goes / shall in the bathtub sit
'Paul will sit / probably sits in the bathtub.'

De Groot (1992) studies Dutch devices of future time reference through questionnaire data in which subjects are asked to verbalize different scenarios. His data corroborate Ten Cate's claim that *gaan* expresses intentions and proximate future events. He also argues that the future meaning of *gaan* depends on the co-occurrence of a dynamic verb. Example (2a) contains a dynamic verbal complement, and therefore allows a future interpretation besides the lexical motion meaning of *gaan*. Example (2b), with a non-dynamic verb, allows only the lexical meaning.

(2) a. Ik ga nu eerst even de brief schrijven. (a-b: De Groot 1992: 4)
I go now first just the letter write
'I am going to write the letter first.'

b. Ik ga een uur in de zon liggen.
I go one hour in the sun lie
'I am going (somewhere) to lie in the sun for an hour.'

The proposed correlation of dynamic verbs with a future interpretation is directly testable through a collexeme analysis. If *gaan* shows a significant preference for distinct groups of dynamic verbs, these verbs should receive a temporal interpretation, rather than express the lexical meaning of self-propelled motion.

Beheydt (2005) compares Dutch *gaan* and English *be going to* in a corpus study based on a set of translated novels. Like Ten Cate and De Groot, she points out the construction is used for future events that are connected to a present cause such as speaker intentions or directly observable evidence (2005: 252). She criticizes earlier approaches such as Haeseryn *et al*. (1997), who report that *gaan* tends to be used to express weather phenomena and changes of state, and that it does not combine with stative verbs. Beheydt argues that weather forecasts are based on present evidence, changes of state require premeditation, and that stative verbs semantically clash with the idea of intentional actions. She concludes that the observed characteristics of *gaan* fall out of its present orientation, and that its exact interpretation is a matter of contextual resolution (2005: 252). For the present analysis, Beheydt's account brings up the question whether semantic frames such as the weather or changes of state have become conventionally associated with the construction, as suggested by Haeseryn *et al.*, or whether they are only incidentally found with *gaan* because of its more general characteristic of expressing future events that connect to the present.

The English future construction with *be going to* has been extensively studied. It was already mentioned here that *be going to* encodes meanings with higher agentivity and dynamicity than the alternative construction with *will* (Gries and Stefanowitsch 2004: 114). An early account of *be going to* is Royster and Steadman (1927), who characterize it as a future construction that conveys the meanings of general intention, earnest purpose, inevitability, and immediacy. These four shades of meaning are illustrated in that order in the examples in (3).

(3) a. Indeed, many essays and books are (a-d: Royster and Steadman 1927: 401)
nothing but explanations of the way in which a writer is going to use a word.
b. He said to himself: I'm going to be an architect.
c. And I am going to fail again as I have failed before.
d. The head of the Oendennises going to marry an actress ten years his senior - the headstrong boy about to plunge into matrimony!

Concerning the diachronic origins of the construction, Royster and Steadman suggest the meaning of intention associated with self-propelled movement gave rise to a sense of immediacy, which then developed into future meaning. This proposal resonates with accounts in Bybee and Pagliuca (1987) and Hopper and Traugott (2003).

Brisard (2001) distinguishes four usage types of *be going to* based on a concordance of 421 examples from the **BROWN** and **LOB** corpora. The first type expresses intended future actions, as illustrated in (4a). Second, Brisard notes that certain examples of *be going to* have the epistemic quality of expressing presupposed information. An example of this is shown in (4b). The example presupposes that the government will provide housing in some way, but leaves open the way in which the task will be accomplished. The third type also conveys epistemic modality. Example (4c) illustrates how *be going to* can encode the inevitability of a future event. Finally, example (4d) presents a future event as imminent.

(4) a. What are you going to do about Sarah?' she asked. (a-d: Brisard 2001: 261-65)
b. It's not for the government to decide how it's going to house people.
c. That tub is going to explode all at once.
d. He looked as if he was going to keel over.

Brisard's account presents finer distinctions of the epistemic meanings of *be going to*. The strongest epistemic interpretation that *be going to* can receive is the presupposition of a future event, as in (4b). The presentation of a future event as inevitable, as in (4c), conveys an epistemic meaning that is still strong, but somewhat weaker. The meanings of imminence and intention do not commit the speaker to the factuality of the future event, as shown in (4d) and (4a). We therefore need to concern ourselves with the question whether the different degrees of epistemic modality map onto a historical development that can be documented through shifting collocational patterns.

Danchev and Kytö (1994) use the **Helsinki Corpus** to analyze the semantic development of *be going to* in a qualitative manner. It is found that examples from Middle English tend to combine the meanings of movement, intention, and proximate future. Among the observed examples up to the 17th century the construction takes infinitive complements such as *bring*, *give*, *meet*, *see*, and *visit* (1994: 69), which allow for interpretations that combine the meanings of movement, intention, and futurity. Danchev and Kytö also observe that the type frequency of the complements of *be going to* increases in Early Modern English (1994: 69). A diachronic distinctive collexeme analysis can be used to interpret such an increase in type frequency in more detail, because it shows what verb types have gained more prominence through this development.

To summarize this survey, the characteristics of Dutch *gaan* and English *be going to* can be contrasted in the following way. Both constructions are said to be biased towards dynamic verbs as preferred complements, but whereas *gaan* is argued to occur with inchoative verbs, *be going to* has been shown to have a preference for telic verbs. A further difference is the purported tendency of *gaan* to express literal motion with stative verbal complements. English examples such as *Bob is going to stay here* do not enforce an interpretation in terms of motion. The following section presents a brief synchronic comparison of the two constructions on the basis of modern corpus data to flesh out these contrasts in some more detail. Any observable differences between the constructions call for an explanation in terms of diachrony: Are the present-day differences between Dutch *gaan* and English *be going to* a recent trend, or are they indicative of separate grammaticalization paths? Little has been said about the diachronic development of Dutch *gaan*, but most studies of English *be going to*

converge in their approval of the hypotheses in Bybee and Pagliuca (1987). Corpus-based studies have noted an increase in type frequency, but it remains to be investigated whether this phenomenon went along with functional changes.

3.1 A brief synchronic comparison

As a prerequisite for the following diachronic analysis, this section aims to determine the contrasts between Dutch *gaan* and English *be going to* in modern usage. Despite the differences outlined above, there arguably are several points of convergence as well. Beheydt (2005) points out that both constructions share an orientation to the present, a preference for intentional or premeditated actions, and a preference for dynamic events. She concludes that '[*G*]*aan* and *be going to* are basically used in the same pragmatic contexts in Dutch and English' (2005: 257). This section compares the collexemes of the two constructions, but restricts its scope to a brief contrast of the ten most strongly attracted collexemes. As a matter of course, the differences between these verbs constitute only the tip of the iceberg, but they are instructive enough to provide a few guiding questions for the subsequent diachronic analysis. Table 11 summarizes the corpus data that is used for the comparison of Dutch *gaan* and English *be going to* in modern usage.

Table 11: Synchronic data for Dutch *gaan* and English *be going to*

CORPORA	SIZE	SEARCH STRINGS	HITS	INFINITIVES
INL	8.28 M	*ga, gaan, gaat*	3,332 (*gaan* + INF)	275,882
BNC	100 M	*going to, gonna*	37,945 (*be going to* + INF)	3,426,582

Two collexeme analyses are performed to determine the most strongly attracted main verb complements for each construction. The ten most strongly attracted collexemes for Dutch *gaan* and English *be going to* are shown in Table 12 below. The sets of collexemes differ markedly with respect to transitivity, lexical aspect, and agentivity.

Table 12: Collexemes of Dutch *gaan* and English *be going to*

DUTCH *GAAN*			ENGLISH *BE GOING TO*	
VERB	GLOSS	COLLSTR	VERB	COLLSTR
regenen	*rain*	73.56	do	Infinite
praten	*talk*	32.80	get	Infinite
gebeuren	*happen*	32.02	happen	Infinite
kosten	*cost*	29.42	say	168.39
waaien	*storm*	26.26	die	125.70
werken	*work*	25.56	cost	93.45
samenwerken	*collaborate*	21.81	put	91.57
zitten	*sit*	18.83	ask	59.91
onderzoeken	*analyze*	18.52	go	58.13
schijnen	*shine*	17.99	marry	52.95

The two sets of collexemes exhibit little overlap in terms of semantically related verbs. The only overlapping elements are Dutch *gebeuren* 'happen' and *kosten* 'cost'; corresponding

English elements are found in the collexemes of *be going to*. Interestingly, these verbs hardly conform to the proposed shared semantics of the two constructions, as they do not orient towards the present, encode intentional actions, or refer to dynamic events.

A first impression of Dutch *gaan* is that it preferably co-occurs with intransitive verbs. The only transitive verb in the ten most strongly attracted collexemes is the verb *onderzoeken* 'analyze'. This verb is not even prototypically transitive, because it encodes an event that is not punctual and that does not strongly affect the patient (Hopper and Thompson 1980). A second characteristic that emerges is that most attracted collexemes encode events that are extended in time. All shown collexemes are imperfective verbs that denote processes which can be extended indefinitely. A third characteristic of the collexemes of *gaan* is their low degree of agentivity. Weather verbs such as *regenen* 'rain' or *waaien* 'storm' do not have an identifiable agent, and verbs such as zitten 'sit' describe human actions that are not very agentive, either. The collexeme analysis corroborates the observation of Haeseryn *et al*. (1997: 975) that *gaan* is typically used in statements about the weather. However, their claim that *gaan* is also used to encode future events that involve a change of state receives little support from the present analysis. Even if changes of state are understood very broadly as situations that are different at their end points than in their initial stages, it must be concluded that *gaan* does not show a strong preference for verbs denoting these situations. The most strongly associated situation type is that of intransitive activities, which covers the group of weather verbs as well as a group of verbs that encode human activities. In the case of weather verbs and verbs such as *gebeuren* 'happen', which select inanimate subject referents, it is clear that human intentions are not at issue and that a modally neutral prediction is being made. Even in examples with *gaan* and verbs that allow human subject referents, the made predictions tend to be relatively free from the meaning of intention.

With regard to transitivity, perfectivity, and agentivity, English *be going to* exhibits the inverse preferences. Among the most strongly attracted collexemes, only *happen*, *die* and *go* do not allow a transitive argument structure. The collexemes *get, say, die, put, ask,* and *marry* denote telic events that are minimally extended in time. These perfective verbs encode events in which a punctual event brings about an abrupt change of state. Lastly, the agentive nature of *be going to* shows itself in the attraction of verbs such as *do, get, say, put, ask, go,* and *marry*. These differences between the two constructions raise the question whether they are due to recent semantic developments in the respective languages, or whether they reflect altogether different paths along which the two constructions grammaticalized. As pointed out in many previous accounts, the meaning of intention figures prominently in the history of *be going to*, and strongly attracted collexemes such as *get, say*, or *marry* speak to the fact that intention still is a major semantic component of the construction in modern usage. With *gaan*, the strong attraction of weather verbs and other non-agentive predicates suggests that intention is not as central to the constructional meaning.

3.2 A diachronic distinctive collexeme analysis of Dutch *gaan*

To assess the semantic development of Dutch *gaan* over time, a diachronic distinctive collexeme analysis is performed on the basis of the non-finite verbal complements occurring with *gaan* in three diachronically ordered text collections that comprise Dutch texts from the Project Gutenberg and the Digital Library of Dutch texts. The relative scarcity of the construction does not allow an analysis in which each century is treated as a different time period, so that two centuries are collapsed into a single period for the earlier data. Table 13 shows the most frequent verbs for each period of time.

Table 13: Top 10 verbs with *gaan* over three periods of time

16-17			18-19			20		
VERB	GLOSS	N	VERB	GLOSS	N	VERB	GLOSS	N
liggen	*lie*	16	zitten	*sit*	38	zitten	*sit*	29
slapen	*sleep*	10	zien	*see*	32	zien	*see*	24
lopen	*walk*	9	slapen	*sleep*	27	slapen	*sleep*	19
zitten	*sit*	7	doen	*do*	26	doen	*do*	14
spreken	*speak*	7	halen	*pull*	24	halen	*get*	11
strijken	*brush*	7	zoeken	*seek*	16	wandelen	*walk*	11
reizen	*travel*	7	opzoeken	*find*	16	liggen	*lie*	10
doen	*do*	6	zeggen	*say*	14	eten	*eat*	10
stellen	*put*	6	liggen	*lie*	12	zoeken	*seek*	9
staan	*stand*	5	leggen	*put*	11	zeggen	*say*	8

The two most frequent elements of the first period are non-dynamic human activity verbs, which are vague with respect to a literal interpretation in terms of movement and a future interpretation. Example (5a) shows that the collocation *gaan slapen* 'go to bed' already existed in the 16th century. Examples (5b) to (5d) illustrate aspectual uses of the posture verbs *liggen* 'lie' and *staan* 'stand'. In these examples, the meaning of the posture verbs is continuative rather than spatial. The denoted event is presented as beginning in the present moment, but temporally extended to the proximate future. Example (5e) shows that already in the 17th century the construction can encode future events just as in present-day Dutch.

(5) a. Daer staet sy d'eerst op, en gaet alderlest slapen. (a-d: GUTENBERG 16-17)
there stands she first up and goes latest sleep
'She gets up the earliest, and goes to bed the latest.'

b. Zij gaet weer zitten drinken
she goes again sit drink
'She will once more sit there drinking.'

c. Ic sal daer binnen gaen ligghen swighen
I shall there inside go lie be.silent
'I shall go inside and stay silent.'

d. soude ic mijnen tijt gaen staen verslijten met schriftuere te preken?
should I my time go stand wear.out with scripture to preach
'Should I keep wasting my time preaching with the Holy Scripture?'

e. 'k Gae trouwen; wilt ghy weten Waerom?
I go marry want you know why
'I'm going to marry, do you want to know why?'

The raw frequencies show that future time reference was not yet the primary function of *gaan* with a non-finite verbal complement in the 16th and 17th century, even if examples with future time reference are attested. The high frequency of movement and posture verbs, which are productive markers of continuative and progressive aspect in Dutch (Haeseryn *et al*. 1997), suggests that the construction with *gaan* was associated with temporally extended events from

an early stage in its grammaticalization. Yet, as was shown in the previous section, movement and posture verbs do not constitute a group of attracted collexemes in present-day usage. A tentative explanation for this is that a continuative aspectual meaning has entered the constructional semantics of *gaan*, so that additional progressive marking by a posture verb would be redundant.

Table 13 indicates that posture and movement verbs stay frequent collocates of *gaan* right until the 20th century, as do idiomatic collocations such as *gaan slapen* 'go to bed'. To see whether these constancies reflect an actual semantic stasis, or whether a development takes place between the 16th and the 20th century, a diachronic distinctive collexeme analysis is performed on 496 different verb types along with their respective frequencies in each period. Table 14 below presents all distinctive collexemes of *gaan* for each of the three periods.

Table 14: Distinctive collexemes of *gaan* over three periods of time

16-17			18-19			20		
VERB	GLOSS	CS	VERB	GLOSS	CS	VERB	GLOSS	CS
lopen	*walk*	3.52	opzoeken	*find*	2.67	beminnen	*love*	3.44
strijken	*run off*	3.30	onderzoeken	*analyze*	2.35	denken	*think*	3.44
stellen	*put*	3.25	vertellen	*tell*	2.25	gebeuren	*happen*	2.45
reizen	*travel*	2.86	doorbrengen	*spend*	2.01	bevrijden	*liberate*	1.96
liggen	*lie*	2.46	geven	*give*	1.68	studeren	*study*	1.96
preken	*preach*	2.00	varen	*travel*	1.68	voelen	*feel*	1.96
treden	*step*	2.00	verkoopen	*sell*	1.68	eten	*eat*	1.66
verhuizen	*move*	2.00	leggen	*put*	1.50	werken	*work*	1.49
leiden	*lead*	1.48	sterven	*die*	1.49	krijgen	*get*	1.47
rechten	*straighten*	1.48	brengen	*bring*	1.49	twijfelen	*doubt*	1.47
spreken	*speak*	1.41	halen	*get*	1.41	kijken	*look*	1.32
drinken	*drink*	1.36	roepen	*call*	1.39			
bezigen	*use*	1.33	nemen	*take*	1.37			

The most distinctive elements of the first period confirm the observation that posture and movement verbs were characteristic of early instances of *gaan*. The posture verb *liggen* 'lie' forms a category with the verbs *stellen* 'put' and *rechten* 'straighten', which denote caused postures. Verbs denoting movement through space are *lopen* 'walk', *strijken* 'run off', *reizen* 'travel', *treden* 'step', *verhuizen* 'move', and *leiden* 'lead'. In addition, the column contains several activity verbs such as *preken* 'preach' or *spreken* 'speak'. With the exception of the caused posture verbs, the distinctive collexemes of this period share an atelic aspectual character. Another common trait is that all distinctive collexemes refer to activities that are commonly done intentionally. The activities of running off, traveling, or using an object usually presuppose a purpose, and hence an intentional agent. The examples with the most distinctive collexemes in the first period thus encode events that involve the intentional movement of an animate agent, as illustrated below. The denoted events lie in the proximate future, or are in their initial stages.

(6) a. Nu willic gaan loopen al in mijn huus. (a-b: GUTENBERG 16)
now want.I go walk all in my house
'Now I want to go home.'

(6) b. Daer gaet hij strijcken!
there goes he run.off
'There he's running off!'

The second period shows a substantial departure from the earlier pattern, as most of the distinctive collexemes have the telic aspectual contour of accomplishment verbs. As in the first period, movement verbs such as *opzoeken* '*find, go to*' and *varen* 'travel', and caused posture verbs such as leggen 'put' are among the most distinctive elements. Many verbs denote the caused motion or transfer of an object, such as *doorbrengen* 'spend', *geven* 'give', *verkoopen* 'sell', *brengen* 'bring', *halen* 'get', and *nemen* 'take'. Examples (7a) and (7b) illustrate how these verbs in connection with *gaan* encode the actions of human agents.

(7) a. Ik houd van avonturen, en ik ga ze opzoeken. (a-c: GUTENBERG 18- 19)
I love adventures and I go them find
'I love adventures, and I'm going to find them.'

b. Nu, myne koets staat gereet; ik ga haar halen
now my carriage stands ready I go her fetch
'Now, my carriage is waiting, I'm going to fetch her'

By contrast, (7c) refers to an event that is unintended. The verb *sterven* 'die' is the only element in the second period that denotes a spontaneous event. Its distinctiveness shows that the constructional meaning is now broadening to accommodate events that are not connected to the intentions of human agents.

(7) c. Uilenspiegel zeide tot Nele: -Liefste nu gaan we sterven.
Uilenspiegel said to Nele darling now go we die
'Uilenspiegel said to Nele: -Darling, now we're going to die.'

This trend is continued in the third period. Here, the distinctive collexemes include the verbs *gebeuren* 'happen' and *werken* 'work', which are also listed as attracted collexemes of *gaan* in Table 12. The examples of *gaan* in 20th century prose thus approximate its general modern usage. The distinctiveness of the verb of occurrence *gebeuren* 'happen' suggests that the future meaning of *gaan* is fully conventionalized in this period. This is shown in example (8a), which clearly resembles parallel modern examples. The list of distinctive collexemes includes several cognitive and emotive verbs, namely *beminnen* 'love', *denken* 'think', *voelen* 'feel', and *twijfelen* 'doubt'. These verbs denote human activities that are often involuntary reactions. Example (8b) illustrates how the activity of doubting is not self-initiated, but a reaction to a stimulus.

(8) a. Wat gaat er dan gebeuren, Sander? (a-b: GUTENBERG 20)
what goes there then happen Sander
'What is going to happen then, Sander?.'

b. en daardoor gaat ge twijfelen aan het echt-zijn dier goedheid
and through.that go you doubt on the true.being of.that benevolence
'and that makes you doubt the reality of that benevolence'

In convergence with the collocational preferences of *gaan* in present-day usage, the distinctive collexemes of the third period are again mostly atelic, with the exception of

bevrijden ‘liberate’. and *krijgen* ‘get’. Movement and posture verbs are no longer among the most distinctive elements.

In conclusion, we can draw the following sketch of the semantic development of *gaan*. Early usages of *gaan* with an infinitive complement commonly refer to events that involve literal and intentional motion. Typical complementing lexical verbs such as *lopen* ‘walk’ or *strijken* ‘run off’ elaborate the manner of the motion event, they do not encode the purpose or goal of the event. The construction becomes associated with atelic situation types, which is further reinforced through the co-occurrence with grammaticalized motion and posture verbs that convey durative and continuative meanings. In its later development, the constructional meaning of *gaan* broadens, such that it also occurs with verbs that have other types of lexical aspect or that are incompatible with the meaning of intentionality. In present-day usage, *gaan* still preferentially occurs with atelic predicates, but intention is no longer a part of the constructional semantics. In the third period of the diachronic study, we find verbs denoting involuntary responses as distinctive elements, and weather verbs as attracted collexemes in the synchronic study. These tendencies do not rule out uses of *gaan* that encode highly intentional telic events such as *Ik ga hem vermoorden* ‘I’m going to kill him’, but they show that sentences of this type do not constitute prototypical uses of the construction.

3.3 A diachronic distinctive collexeme analysis of English *be going to*

In this section, a diachronic distinctive collexeme analysis is performed to investigate the development of English *be going to* from the early 18th century up to the 20th century. The present study uses the **Corpus of Late Modern English Texts** (CLMET), which covers three successive periods of time from 1710 to 1920; the OED is used as a supplement. Since all quotations in the OED are tagged with a publication date, it is possible to match the retrieved examples with the respective periods of the CLMET. Table 15 shows for each period the ten most frequent verbs that co-occur with *be going to*. The frequencies show strong similarities between the first two periods, while the third period displays a markedly different pattern.

Table 15: Top 10 verbs with *be going to* over three periods of English

1710-1780		1780-1850		1850-1920	
VERB	N	VERB	N	VERB	N
say	12	say	21	be	128
fight	8	make	16	have	64
give	8	tell	16	do	63
make	7	give	14	say	43
begin	6	be	14	make	42
tell	6	marry	12	take	39
marry	5	have	12	get	35
send	5	leave	12	leave	27
answer	4	take	12	give	25
observe	4	see	11	tell	22

The verbs *say*, *give*, *make*, *tell*, and *marry* overlap in the first two periods, suggesting that the meaning of *be going to* under these periods strongly harmonizes with telic and dynamic verbs. In the second period, the stative verbs *be* and *have* have entered the list of the ten most frequent items, and by the third period, they instantiate the two most frequent complement

types. The most frequent types in the third period actually resemble a general list of the most frequent verbs in English, with *be*, *have*, and *do* at the top, and *say*, *make*, *take*, and *get* within the top ten. This suggests that the construction has become more widely applicable, and that its meaning has become more general and schematic, in a word, more grammaticalized. Table 16 shows for each period the distinctive collexemes of *be going to*. In the second period, only six elements are judged to be distinctive.

Table 16: Top 10 distinctive collexemes of *be going to* over three periods of English

1710-1780		1780-1850		1850-1920	
VERB	COLLSTR	VERB	COLLSTR	VERB	COLLSTR
fight	3.48	hunt	2.31	be	10.66
publish	2.32	speak	2.21	do	4.81
answer	1.94	commence	1.79	get	4.18
observe	1.94	expose	1.79	have	3.19
embrace	1.92	part	1.79	try	2.94
ravish	1.92	strike	1.79	die	2.30
relate	1.69			stay	1.76
begin	1.56			happen	1.74
visit	1.48			run	1.57
				talk	1.37

All distinctive collexemes of the first period select for animate, intentional subject referents. The most distinctive element is *fight*, which is a highly intentional verb. Example (9a) shows that uses of *going to fight* can still convey a literal sense of motion, which is no longer found in modern usage. Among the other distinctive elements for the first period, the verbs *answer*, *observe*, and *relate* are meta-linguistic verbs, which denote intentional speech acts. As example (9c) shows, even *begin* tends to be used with meta-linguistic meaning.

(9) a. And now my boy, I cried, you are going to fight for your country. (CLMET)
b. By the circumstances of the story which I am going to relate,
you will be convinced of my candour.
c. As he was going to begin his narrative, Rasselas was called to a concert.

In the second period, only six elements are judged to be significantly distinctive. Like the elements in the first period, all of these are compatible with an intentional reading. Speech act verbs continue to be of importance to the construction, as evidenced by the distinctive collexeme *speak*, and uses of *commence* like example (10b). Example (10c) shows that despite its preference for intentional future actions, the construction is used to convey imminent future events that do not depend on human agents.

(10) a. 'We're going to hunt Boney out, Sir,' Dobbin said. (CLMET)
b. The orator had finished one story, and was going to commence another.
c. In the true sleepy tone of a Scottish matron when (OED)
ten o'clock is going to strike.

In the third period, the highly frequent elements *be*, *do*, and *have* actually turn out to be also among the most distinctive elements. The high ranking of these elements suggest an increased generalization of the construction. Increased usage of schematic verbs such as *be* and *do*, as

illustrated in examples (11a) and (11b), is indicative of a more general applicability of the construction.

(11) a. There is going to be some serious trouble here, I'll lay my last dollar on that. (a-b: CLMET)
b. "What are you going to do?" asked George's father.

Out of the list of ten distinctive collexemes, the verbs *do*, *get*, *die*, and *happen* appeared earlier in Table 12 as attracted collexemes of *be going to* in modern usage. The fact that the verb *happen*, which did not appear as one of the most frequent elements for the third period, is judged as distinctive for this stage corroborates the hypothesis that the occurrence of spontaneous, non-intended events is only encoded by *be going to* in later stages of its development. The expression of this type of future events is illustrated in the below examples with *have*, *die*, and *happen*.

(11) c. "Are we going to have an accident, Uncle Swithin?" (c-e: CLMET)
d. In his small stock of knowledge, he knew, like all around him, that he was going to die.
e. Carrie was particularly excited, and said she hoped nothing horrible was going to happen.

A difference between the attracted collexemes of *be going to* in present-day English and the distinctive collexemes of the third period is that the latter do not exhibit the strong preference for achievement verbs that was observed in the modern data.

3.4 Results and discussion

The present analysis has compared Dutch *gaan* and English *be going to* in their synchronic usage and their historical development. A first result of the investigation is that the two constructions differ substantially in their collocational preferences in modern usage. This finding stands in contradiction to accounts that have stressed the similarities between Dutch *gaan* and English *be going to*. While *gaan* and *be going to* exhibit some common traits, the two constructions differ considerably on the parameters of transitivity, perfectivity, and agentivity. Dutch *gaan* attracts verbal complements that are intransitive, temporally extended, and non-agentive; and English *be going to* attracts verbs that are transitive, punctual, and highly agentive. The investigation of diachronic corpus data suggests that these differences are not due to recent semantic changes. Instead, the developments of the two constructions have proceeded in the context of different collocating verbs.

In its early usages, Dutch *gaan* commonly co-occurred with motion verbs such as *lopen* 'walk'. Crucially, motion verbs are typically imperfective. The activity of movement is internally homogeneous and can be extended indefinitely. In later usage, the constructional meaning broadens. The construction accommodates verbs without the meaning of movement and intentionality. In present-day usage, intention is no longer a part of the constructional semantics of *gaan*, but the construction still preferentially occurs with atelic predicates. Hence, we find cognitive response verbs such as *denken* 'think' and *voelen* 'feel' in the third historical period, and weather verbs such as *waaien* 'storm' as attracted collexemes in the synchronic study.

Also English *be going to* is used to encode intentional movement in its early usages, as is evidenced by its early collexemes. However, unlike Dutch *gaan* it never had a preference for motion verbs as infinitive complements. Instead, speech act verbs, as prototypical intentional verbs, are central to its development. Verbs such as *answer* or *begin* are perfective, and thus prefigure the modern tendency of *be going to* to express punctual, telic events. The diachronic distinctive collexeme analysis further shows that the development to future meaning is accompanied by a growing preference for general, light verbs. This suggests that the construction became more general and applicable to a wider array of contexts.

To conclude, while English *be going to* and Dutch *gaan* both follow the general path of motion-based futures (Bybee et al. 1991: 32), this does not mean that they function similarly in language use. Converse preferences for perfectivity, transitivity, and agentivity can be shown to permeate their respective developments. A historical perspective on the shifting collocational preferences of the two constructions reveals that *be going to* had a special affinity towards speech act verbs, while with *gaan*, movement verbs had a special role.

4 Concluding remarks

Diachronic collostructional analysis is one way of studying the development of grammatical constructions. If suitable corpora are available, shifting collocational patterns will likely reveal some semantic developments that would otherwise remain undetected. If these developments can be meaningfully tied to previous research, then diachronic collostructional analysis even provides an empirical way to engage in theoretical discussions. That said, constructions of course change with respect to many parameters, collocates being merely one of them; and sometimes differences between the distinctive collexemes of historical periods may be difficult to interpret. I hope to have shown though that it can be worth the effort and that it can substantially deepen our understanding of how grammatical constructions change.

5 Appendix: Some practical issues

Reading a paper such as this one can give you a general idea of how collostructional analysis works, but there really is no better way of learning about it than trying it out yourself. Stefan Gries (stgries@linguistics.ucsb.edu, http://www.linguistics.ucsb.edu/faculty/stgries) has written a script for the open-source statistical package R that can be used for all techniques described in this paper. Stefan will provide the script upon request. This may be a good opportunity to start working with R, which is really a very practical tool for many different purposes. The documentation of the script includes sample input and output files, as well as references to more literature.

What corpora should you use? This largely depends on what you want to do. Generally, a good place to go is Mark Davies' on-line interface of the British National Corpus (BNC), (http://corpus.byu.edu). The site also has two diachronic corpora, the Corpus of Contemporary American English (COCA), and the TIME magazine corpus. For some collostructional analyses, it is actually more practical to have the entire corpus on your hard disk, as online interfaces usually have some sort of limitation on the number of search results or search syntax. This can be bothersome if, for instance, you would like to determine how many instances of the verb *to be* occur in the corpus you are using. So check what corpora your university library makes available. A good website that presents an overview of corpora and corpus-linguistic tools is David Lee's webpage (http://devoted.to/corpora). Corpora on your

hard disk can be searched through a concordance program such as AntConc, which is freely available for all platforms (http://www.antlab.sci.waseda.ac.jp/software.html).

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